DINUJA PERERA

+44 730 713 8797 | United Kingdom |dinujakdr@gmail.com | linkedin.com/in/dinuja-perera | Portfolio:dinuja.com

PROFESSIONAL SUMMARY

Machine Learning Engineer with 2+ years of experience developing supervised, unsupervised, and deep learning models. Research experience in NLP and time series forecasting. Hands-on experience with Azure Databricks, object-oriented Python, GPU computing, version control, and agile model development. Holder of a Tier 1 Global Talent visa, endorsed by UK Research and Innovation (UKRI), with full UK work rights, no sponsorship required, and open to relocation.

Key Skills:

- **AI/ML**: Python, TensorFlow, Keras, PyTorch, Scikit-learn, LSTM, ESN, RNN, Random Forest, SVM, KNN, K-Means, NLP, Transformers (BERT, RoBERTa), Isolation Forest, Autoencoders
- **Data & Pipelines**: PySpark, Pandas, SQL, DuckDB, MongoDB, Cassandra, Redis, Neo4j, Azure Databricks, scalable data extraction pipelines
- **Agentic AI & LLMs:** LangChain (LangChain London Hackathon team member, LangChain Foundation Course), active LangChain community member
- **Collaboration**: Agile, ClickUp, Confluence, Power BI, Tableau, cross-functional stakeholder engagement

WORK EXPERIENCE

KTP Associate – Machine Learning Engineer (Predictive Maintenance) | Uniper UK | Loughborough University - June 2024 - present

- Developed LSTM and ESN time-series models on ~200 turbine sensor streams to predict outputs across multiple load conditions, achieving under 3% average prediction error — directly applicable to Nomad Digital's condition-based fleet monitoring mission.
- Built ML models for sensor health monitoring: used correlation-threshold methods (Spearman + Fisher Z transformation) to distinguish faulty sensor behaviour from normal operating variability, enabling proactive fault detection.
- Designed a scalable missing-value strategy using null-percentage rules, sensor-family imputation, Random Forest regression, and flag columns to preserve full time-series integrity across fixed model inputs.
- Engineered scalable data extraction pipelines, resolving WebSocket timeouts during 2h40m server pulls and reducing sensor lookup complexity from O(n) to O(1) via hash-based indexing.
- Applied K-Means clustering to identify distinct turbine rundown behaviour patterns, supporting operational decision-making.
- Version-controlled all datasets, models, and experiments in GitHub; maintained structured documentation for reproducibility and knowledge transfer.
- Collaborated with reliability engineers, plant operators, academic supervisors, and government stakeholders to translate operational requirements into ML solutions.

Machine Learning Engineer| CML Insight Inc., Texas, USA | Nov 2021 – Jan 2023

- Engineered NLP features (tokenisation, embeddings, sentiment scores) and developed cohort-based models to analyse behavioural patterns across customer groups.
- Maintained end-to-end ML pipelines to enable model reuse on incoming data; practiced OOP Python for code organisation and reusability.
- Assessed feature importance and target leakage to improve model reliability; onboarded and mentored interns on ML workflows.

Teaching Assistant | Department of Information Systems Engineering, University of Colombo, Sri Lanka | Jun 2019 - Nov 2021

- Delivered practical sessions and supervised projects for MSc/BSc students in Machine Learning & Neural Computing, Data Analytics, and Embedded Systems.

Intern Data Analyst| Brandix Apparel Limited |Jun 2019

- Analysed a large spare parts dataset, using Microsoft Excel Macros and Pivot Tables to detect data discrepancies and inaccuracies.
- Collaborated with cross-functional teams to address data quality issues and implement corrective actions.

Research Studies/ Projects:

- **Term Deposit Subscription Prediction and Customer Segmentation:**
  Built a two-stage ML pipeline to predict bank customers likely to subscribe to term deposits using demographic and campaign data.
  Reduced unnecessary campaign calls, saving 1,693 staff hours (212 working days) across 40,000 customers.
  Segmented subscribers into 6 clusters using KMeans to guide call timing, campaign strategy, and customer engagement.
- **Customer Dissatisfaction Prediction in Logistics Delivery:**
  Built a model to identify drivers of customer dissatisfaction in logistics surveys.
  Achieved **0.77 recall for unhappy customers** using ensemble methods.
  Identified key factors affecting satisfaction, informing the service improvement and survey redesign.

- **Leveraging NLP for Enhanced Company Performance: Transformer-Based Sentiment Analysis and Comparative Review -** Implemented transformer-based sentiment analysis alongside topic modeling and aspect-based analysis, to provide actionable insights for companies to improve products, services, and organisational aspects.
- **Benchmark NLP Algorithm for Hate Speech Detection on Social Media -** Developed 12 deep learning architectures, including RNNs, CNNs, BERT, RoBERTa), and hybrid architectures (e.g., CNN + LSTM), employing a three-step approach involving basement establishment, hate speech detection using custom models, and addition of oppositeness.
- **Comparative NLP model analysis on Hate Speech Detection on Social Media Platforms -** Demonstrated RoBERTa's superior performance over BART in both binary and multi-class classification tasks.

EDUCATION

MSc Data Science - Distinction | University of Greenwich, London | 2023 – 2024

- Modules: Data Visualisation, Statistical Methods for Time Series Analysis, Graph and Modern Databases, Big Data, Blockchain for FinTech Applications.
- Completed projects involving comparative model evaluation (regression, classification, neural networks), clustering analysis, and optimisation of models for diverse datasets.

MSc specialising in Data Science, Analytics and Engineering| University of Moratuwa, Sri Lanka | 2021

- Modules: Pattern Recognition, Data Mining, Business Intelligence, Neural Networks, Algorithms, Statistical Inference, Bioinformatics.

Grad. Dip. in Electronics Telecommunication & Computer Engineering - Distinction| Institution of Engineers, Sri Lanka | 2020

- Attained a GPA of 3.75 (/4.20) in modules including Software Engineering Practices, Computer Security, Computer Networks.

BSc (Electronics & IT Theme) - Second Class Honours| University of Colombo, Sri Lanka | 2015-2019

- Earned a GPA of 3.24 (/4) in modules including Applied Mathematics, Statistics, Computer Science, and Physics.

PUBLICATIONS / CONFERENCE PRESENTATION

"Short Comparative Analysis on Pretrained BART and RoBERTa in Detecting Hate Speech on YouTube and Reddit Platforms," presented at WiNLP Workshop co-located with EMNLP 2022.

EXTRACURRICULAR & CERTIFICATIONS

- Leadership & Management - CMI Level 5 Equivalent - Ashorne Hill Management College, UK
- LinkedIn Learning Certifications: Advanced NLP, PySpark for Big Data, Azure DevOps Fundamentals, Generative AI Foundations